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Week-03-02-Practice Session-Coding: Attempt review | REC-CIS
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Write a program that determines the name of a shape from its number of sides. Read the number of sides from the user and then report the appropriate name as part of a meaningful message. Your program should support shapes with anywhere from 3 up to (and including) 10 sides. If a number of sides outside of this range is entered then your program should display an appropriate error message.

Sample Input 1

3

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     if(n==3)
7     {
8         printf("Triangle");
9     }
10    else if(n==4)
11    {
12        printf("Square");
13    }
14    else if(n==5)
15    {
16        printf("Pentagon");
17    }
```

```
19 {  
20     printf("Hexagon");  
21 }  
22 else if(n==7)  
23 {  
24     printf("Heptagon");  
25 }  
26 else if(n==8)  
27 {  
28     printf("Octagon");  
29 }  
30 else if(n==9)  
31 {  
32     printf("Nonagon");  
33 }  
34 else if(n==10)  
35 {
```

```
30     else if(n==9)
31     {
32         printf("Nonagon");
33     }
34     else if(n==10)
35     {
36         printf("Decagon");
37     }
38     else
39     {
40         printf("The number of sides is not supported.");
41     }
42 }
```

	Input	Expected	Got
✓	3	Triangle	Triangle
✓	7	Heptagon	Heptagon
✓	11	The number of sides is not supported.	The number of sides is not support

Passed all tests! ✓

The Chinese zodiac assigns animals to years in a 12-year cycle. One 12-year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the Dragon, and 1999 being another year of the Hare.

Year	Animal
2000	Dragon
2001	Snake
2002	Horse

```
1  #include<stdio.h>
2  int main()
3  {
4      int year;
5      scanf("%d",&year);
6      if (year%12==8)
7      {
8          printf("Dragon");
9      }
10     else if(year%12==9)
11     {
12         printf("Snake");
13     }
14     else if(year%12==10)
15     {
16         printf("Horse");
17     }
```

```
16     printf("Horse");
17 }
18 else if(year%12==11)
19 {
20     printf("Sheep");
21 }
22 else if(year%12==0)
23 {
24     printf("Monkey");
25 }
26 else if(year%12==1)
27 {
28     printf("Rooster");
29 }
30 else if(year%12==2)
31 {
32     printf("Dog");
33 }
```



```
33     }
34     else if(year%12==3)
35     {
36         printf("Pig");
37     }
38     else if(year%12==4)
39     {
40         printf("Rat");
41     }
42     else if(year%12==5)
43     {
44         printf("Ox");
45     }
46     else if(year%12==6)
47     {
48         printf("Tiger");
49     }
```

```
44     printf("ox");  
45 }  
46 else if(year%12==6)  
47 {  
48     printf("Tiger");  
49 }  
50 else  
51 {  
52     printf("Hare");  
53 }  
54 return 0;
```

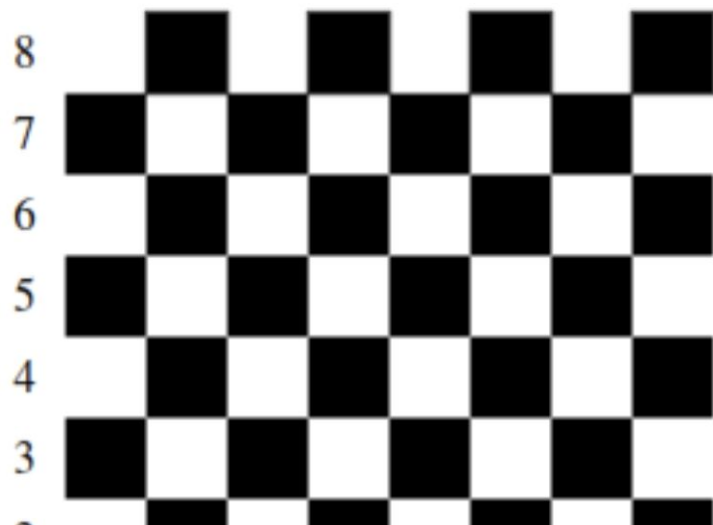
	Input	Expected	Got	
✓	2004	Monkey	Monkey	✓

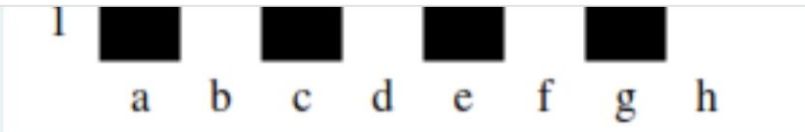
```
52     printf("Hare");  
53 }  
54 return 0;
```

	Input	Expected	Got	
✓	2004	Monkey	Monkey	✓
✓	2010	Tiger	Tiger	✓

Passed all tests! ✓

Positions on a chess board are identified by a letter and a number. The letter identifies the column, while the number identifies the row, as shown below:





Write a program that reads a position from the user. Use an if statement to determine if the column begins with a black square or a white square. Then use modular arithmetic to report the color of the square in that row. For example, if the user enters a1 then your program should report that the square is black. If the user enters d5 then your program should report that the square is white. Your program may assume that a valid position will always be entered. It does not need to perform any error checking.

Sample Input 1

a 1

```
1 #include<stdio.h>
2 int main()
3 {
4     int num,sum;
5     char alpha;
6     scanf("%c %d",&alpha,&num);
7     sum=alpha+num;
8     if(sum%2==0)
9     {
10         printf("The square is black.");
11     }
12     else
13     {
14         printf("The square is white.");
15     }
16 }
```



	Input	Expected	Got	
✓	a 1	The square is black.	The square is black.	✓
✓	d 5	The square is white.	The square is white.	✓

Passed all tests! ✓

month, then scan across to the appropriate month column and read the day of year number. Reverse the process to find the standard date for a given day of year.

Write a program to print the Day of Year of a given date, month and year.

Sample Input 1

18

6

2020


```
1  #include<stdio.h>
2  int main()
3  {
4      int i,mon,yr,inday;
5      scanf("%d%d%d",&inday,&mon,&yr);
6      char daypm[]={31,28,31,30,31,30,31,30,31,30,31,30};
7      int day=inday;
8      if((yr%4==0&&yr%100!=0)|| (yr%400==0))
9      {
10         daypm[1]=29;
11     }
12     for(i=0;i<mon-1;i++)
13     {
14         day=day+daypm[i];
15     }
16     printf("%d",day);
17     return 0;
```



	Input	Expected	Got	
✓	18 6 2020	170	170	✓

Passed all tests! ✓